Strong unextendibility of small Pauli MUB sets: exact bounds for two, three and four qubits

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Abstract

A set of mutually unbiased stabiliser bases (Pauli MUBs) is *weakly unextendible* if no further Pauli class is unbiased to all of them, and *strongly unextendible* if no vector at all is. For a set $\mathcal{S}$ in dimension d we study $\mathcal{F}(v) = \sum_e \big(|\langle e|v\rangle|^2 - |v|^2/d\big)^2$, the sum over all basis vectors e of the set; strong unextendibility is $\min_{|v|=1} \mathcal{F} > 0$. We prove exact, quantitative lower bounds. For two qubits, the minimum is exactly $1/4$. For three qubits, all 24 192 five-element sets form one Clifford orbit and satisfy $\mathcal{F} \geq (1/8 - 10^{-6})|v|^4$, while stabiliser states attain $1/8$. For four qubits, the smallest weakly unextendible sets have exactly five elements (an exhaustive search, matching Grassl's size $q^{n/2}+1$). There are 105 283 584 of them in two types. One type satisfies $\mathcal{F} \geq |v|^4/80$ exactly, so five bases are strongly unextendible in $d = 16$, and its minimum (≈ 0.0524) lies below every stabiliser value. The proofs combine a twirl identity, the graph-theoretic uncertainty relation of de Gois, Hansenne and Gühne, Pauli-symmetry reduction of sum-of-squares certificates, facial reduction, and verification in exact rational arithmetic.

1 Setting

Let n qubits carry the Pauli group with symplectic space $W(2n-1, 2)$. Its points are Pauli operators up to phase, and its Lagrangians are maximal commuting classes. Disjoint Lagrangians give mutually unbiased stabiliser bases, and a weakly unextendible set is a maximal partial spread [1, 2]. Mandayam, Bandyopadhyay, Grassl and Wootters [1] constructed such sets with $d/2+1$ elements for $d = 4, 8$. They proved strong unextendibility for specific examples with a Gröbner basis computation, and noted that a general proof was open. Kantor [3] surveys maximal symplectic partial spreads and reports Grassl's sets of size $q^{n/2}+1$.

Twirl identity. For a stabiliser basis B_L, $\sum_{e\in B_L} |e\rangle\langle e|^{\otimes 2} = d^{-1}\sum_{P\in L} P^{\otimes 2}$. Hence, with C the covered Paulis and U the uncovered ones,

$$\mathcal{F}(v) = \frac{1}{d}\sum_{P\in C}\langle P\rangle^2 = \frac{1}{d}\Big(d - 1 - \sum_{P\in U}\langle P\rangle^2\Big) \qquad (|v| = 1), \tag{1}$$

since $\sum_{P\neq I}\langle P\rangle^2 = d-1$ for pure states. A stabiliser state of a Lagrangian meeting C in m points gives $\mathcal{F} = m/d$. Maximality forces $m \geq 1$, so $\min \mathcal{F} \leq m^*/d$, where m^* is the least such m.

Lovász bound. De Gois, Hansenne and Gühne [4] prove $\sum_i \langle A_i\rangle^2 \leq \vartheta(G)$ for dichotomic observables, where G is their anticommutation graph and ϑ the Lovász number [5]. Applied to U, $\min \mathcal{F} \geq (d - 1 - \vartheta)/d$. The bound is tight when $\vartheta = \alpha(G) = d - 1 - m^*$.

2 Results

Theorem 1 (two qubits)**.** *For every three-element maximal partial spread of* $W(3,2)$*,* $\min \mathcal{F} = 1/4$ *exactly. Here* $\vartheta = \alpha = 2$*, and equivalently the six uncovered Paulis split into two pairwise-anticommuting triples.*

Theorem 2 (three qubits)**.** *The maximal partial spreads of* $W(5,2)$ *have sizes 5 (24 192 sets, one* $\mathrm{Sp}(6,2)$*-orbit) and 9 (complete sets). For every five-element set and every* $v \in \mathbb{C}^8$*,* $\mathcal{F}(v) \geq (1/8 - 10^{-6})|v|^4$*, and stabiliser states attain* $1/8$*. A weaker but structural bound is* $\mathcal{F} \geq \frac{3-\sqrt{5}}{8}|v|^4$*, from* $\vartheta \leq 4 + \sqrt{5}$*, certified over* $\mathbb{Q}(\sqrt{5})$ *in the rank-21 coherent closure of the uncovered graph.*

Method. $\mathcal{F}(v)|v|^2 - \delta|v|^6$ is written as an SOS in the degree-3 monomials and in the monomials $v_i v_j \bar{v}_k$. The Pauli group acts on both monomial spaces with centre -1, so each is a multiple of the unique 8-dimensional irrep, and invariant Gram matrices reduce to blocks of size 15 and 36. All coefficients are Gaussian rationals. At $\delta = 1/8 - \varepsilon$, a strictly interior certificate is rounded (denominator 16). Its pivot variables are restored exactly, every coefficient equation is verified in integer arithmetic, and positivity is checked by exact LDL^{T}, for $\varepsilon = 10^{-3}, \ldots, 10^{-6}$. At $\delta = 1/8$ the 160 stabiliser minimisers force the 15-block to vanish, and facial reduction leaves a 16-dimensional face with zero interior margin. This suggests that the degree-6 bound approaches $1/8$ without attaining it.

Theorem 3 (four qubits)**.** *(i)* $W(7,2)$ *has no maximal partial spread of size 3 or 4. The smallest size is 5, and there are* 105 283 584 *such sets. (ii) They fall into two invariant types. For a set of the first type (lines listed in the repository),* $\vartheta \leq 74/5$*, so* $\mathcal{F} \geq |v|^4/80$ *exactly, while numerically* $\min \mathcal{F} = 0.0523626 < 1/16$*. Sets of the second type are* lifts *of the three-qubit five-sets. For each of them, 120 uncovered points* p *project the set onto a maximal five-set of* $p^{\perp}/\langle p \rangle \cong W(5,2)$*, whereas the first type has no such point. They inherit a mirror partner (a second partition of the same covered set) and have numerical minimum* $1/16$*. (iii) For several inequivalent nine-element sets,* $\min \mathcal{F} = m^*/16$ *exactly (*$1/16$ *or* $1/8$*), via a facially reduced Lovász dual.*

Method for (i). By Witt's theorem, $\mathrm{Sp}(8,2)$ is transitive on ordered pairs of complementary Lagrangians, so every candidate can be taken to contain a fixed pair. The remaining one or two Lagrangians are enumerated exhaustively. As a control, the same code finds no sizes 3 and 4 for three qubits, and finds 56 five-sets through a fixed pair, matching $24\,192 \cdot 20/(135 \cdot 64) = 56$. *Method for (iii).* The maximum independent sets of the uncovered graph are the Lagrangians meeting C in m^* points, minus those points. Complementary slackness with them gives linear equations on the dual. Imposing these equations makes rounding succeed, and the certificate is checked by exact LDL^{T}.

Remark (two qutrits). For the 432 five-element maximal partial spreads of $W(3,3)$, $\mathcal{F} \geq |v|^4/9$ exactly, via an S_6/Jucys–Murphy identity. The true minimum is numerically $(207 - 19\sqrt{57})/432 \approx 0.14711$, which is an identification, not a proof. Mirror partners there lie in opposite orbits and are exchanged only by antiunitaries; a third-moment witness of size $16/27$ separates them. In characteristic 2 the corresponding partners are Clifford-equivalent.

3 Summary and open questions

d	smallest size	$\min \mathcal{F}$	status
4	3	$1/4$	exact
8	5	$\in [1/8 - 10^{-6}, 1/8]$	exact interval
9	5	$\geq 1/9$; ≈ 0.14711	exact bound; numerical value
16	5	$\geq 1/80$; ≈ 0.05236 (one type)	exact bound; numerical value

Open: an exact certificate at $1/8$ (degree 8, or a structural proof); closed forms for the non-stabiliser optima; the strong unextendibility constant for all five-element sets in $d = 16$ and beyond. *Not claimed*: novelty of the weak-unextendibility census at $d = 16$, which may already be in Grassl's tables [3]. All certificates are in the repository (`data/*.json`) with the scripts that verify them.

References

[1] P. Mandayam, S. Bandyopadhyay, M. Grassl, W. K. Wootters, Unextendible mutually unbiased bases from Pauli classes, Quantum Inf. Comput. 14 (2014); arXiv:1302.3709.

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[3] W. M. Kantor, On maximal symplectic partial spreads, Adv. Geom. (2017).

[4] C. de Gois, K. Hansenne, O. Gühne, Uncertainty relations from graph theory, Phys. Rev. A 107, 062211 (2023).

[5] L. Lovász, On the Shannon capacity of a graph, IEEE Trans. Inf. Theory 25 (1979).